Jimmy Hickey, PhD

jimmyjhickey@gmail.com | jimmyjhickey.com | /in/jimmyjhickey | (919) 480 2225

Skills

Languages: Python, R, Julia, C/C++, Bash/Unix, Java, SQL

Tools & Frameworks: PyTorch, TensorFlow, CUDA, Linux, HPC Environments, Git

Methods: Transfer Learning, Generative AI, Machine Learning, Uncertainty Quantification, NLP/LLMs

Soft Skills: Statistical Consulting, Teaching, Data Visualization, Interdomain Communication

Professional Experience

Machine Learning Engineer 2024 - present
Optum

- Use a combination of traditional machine learning methods and LLMs to develop an end-to-end automated auditing pipeline with more accurate results than human auditing.
- Practice modular design to minimize coding time across different disease diagnoses.
- Design flexible diagnosis modules to handle ambiguous medical specifications.
- Create a generalized unit test framework to decrease debugging time for the team.

Statistical Scientist 2020 - 2024
Sandia National Laboratories

- Solved problems involving combining multiple remote sensing data sets, physics based statistical modeling, and computer vision.
- Optimized statistical computing methods, reducing computational complexity by a factor of $O(n^2)$.
- Supported national defense using RNNs and deep learning for signal processing.
- Modeled trajectories for engineering applications using functional data analysis.

Co-Founder 2020 - 2023
NCSU Statistics in the Community

- Secured clients for a pro bono consulting group.
- Visualized data and predictive results in reports and presentations given to non-statistical clients.
- Manage and mentor new consultants.
- Mapped unmet need to allocate donated furniture and analyzed school system data for unmet adolescent need. [report]
- Developed interpretable metrics for donation lag times. [report]
- Analyze newsletter data to improve click-through and subscriber retention. [report]

Machine Learning Researcher 2019 - 2024
Duke Clinical Research Institute

- The applied focus of this work has been to improve stroke risk prediction. Using a combination of multiple studies, we were able to successfully improve stroke diagnosis, especially in underrepresented minorities. The goal of this work is to improve patient care and equity.
- Lead development and publication of a flexible machine learning method improving predictive performance and interpretability of time-to-event outcomes (see publications 2).
- Develop novel transfer learning methods addressing demographic bias in predictive modeling (see publication 3).

Statistics Researcher 2019 - 2024
North Carolina State University

- The applied focus of my work has been to improve patient care by looking at medical data. We have been successfully able to improve predictions of whether patients will go into shock upon being admitted into the ICU and to predict common dental measurements that can be early signs of tooth decay.
- Create a source free Bayesian transfer learning method improving predictive performance with uncertainty quantification (see publications 1 and 4).
- Implement state-of-the-art transfer learning methods in Python, R, and Julia.
- Implement Generative AI models to compare method performance (see publication 6).
- Present work at conferences and events to diverse audiences.

Tutor 2016 - 2022
Winona State University & North Carolina State University

- Tutored students in computer science, statistics, physics, and mathematics.
- Assisted the highest number of students per year at Winona State University.
- Provided one-on-one and small group tutoring, and classroom teaching.

Genomic Systems Programmer Analyst 2018 - 2019
Mayo Clinic

- Developed variant annotation and microbiome pipelines for researchers and clinicians.
- Optimized bioinformatics algorithms to parallelize for high-performance computing (HPC).
- Create onboarding procedures for new hires.

Software Engineer 2016 - 2018
Digi International

- Build firmware for microcontrollers, routing devices, and embedded systems.
- Write automated unit, integration, and system tests for multiple teams.
- Participate in an agile development environment.

Publications

1. **J Hickey**, J P Williams, E C Hector (2024). Transfer Learning with Uncertainty Quantification: Random Effect Calibration of Source to Target (RECaST). *The Journal of Machine Learning Research* [manuscript]
2. **J Hickey**, R Henao, M Pencina, D M Wojdyla, M Engelhard (2024). Adaptive Discretization for Event PredicTion (ADEPT). *AISTATS* [manuscript]
3. C Hong, M Liu, D M Wojdyla, **J Hickey**, M Pencina, R Henao (2024). Trans-Balance: Reducing Demographic Disparity for Prediction Models in the Presence of Class Imbalance. *The Journal of Biomedical Informatics* [manuscript]
4. **J Hickey**, E C Hector, J P Williams (202x). Multivariate and Online Transfer Learning with Uncertainty Quantification. *In Review at Statistics in Medicine* [arXiv]

Education

Ph.D. Statistics 2024
M.S. Statistics 2020
North Carolina State University
Transfer Learning and Survival Analysis Methods with Biomedical Applications

B.S. Computer Science; B.S. Physics; B.A. Mathematics 2018
Winona State University
Minors: Statistics, Data Science